Lectures 7-8 continued

Bivariate random variables: independence and conditional distributions.

Independence of random variables

Definition Two random variables X and Y are independent if

$$P(X \in A \text{ and } Y \in B) = P(X \in A) \cdot P(Y \in B)$$

for **every** two sets $A \subseteq \mathbb{R}$ and $B \subseteq \mathbb{R}$.

Theorem 1 Assume $\begin{cases} (X,Y) \text{ have c.d.f } F. \\ X \text{ has cdf } F_X \\ Y \text{ has cdf } F_Y \end{cases}$

X and Y are independent *if and only if*

$$F(x,y) = F_X(x) \cdot F_Y(y), \text{ for}$$

all $x, y \in \mathbb{R}$.

Theorem 2 (Corollary 3.5.1 + comments following it).

Two random variables X and Y are independent **if and only if**

$$f(x,y) = f_X(x) \cdot f_Y(y), \text{ for all}$$

real numbers x and y.

Note
- $f(x,y)$ is the joint p.f. if (X,Y) is discrete; f_X and f_Y are the p.f. marginals
- $f(x,y)$ is the joint p.d.f. if (X,Y) is continuous; f_X and f_Y are the p.d.f. marginals.

IMPORTANT if we believe that the distribution of X does not change when we learn the value of Y (and vice-versa), we can **model** X and Y as independent.

- Example 3.5.7 and Table 3.5 show how to construct the p.f. of (X, Y) from the p.f.'s of X and Y, since X and Y can be assumed independent.

Note For (X, Y) with joint p.f. given by a (contingency) table we have: X and Y are independent iff the rows in the table are proportional with one another.

- Example 3.5.9

X = Rain in May, 2010

Y = Rain in May, 2011.

$\mathbb{P}(X + Y \geq 1) = ?$

- It is reasonable to assume that X and Y are independent.
- Say that we know the p.d.f of X and Y, and say that they are given by the same function:

$$f_X(x) = \begin{cases} 2x; & 0 \le x \le 1 \\ 0; & \text{otherwise} \end{cases}$$

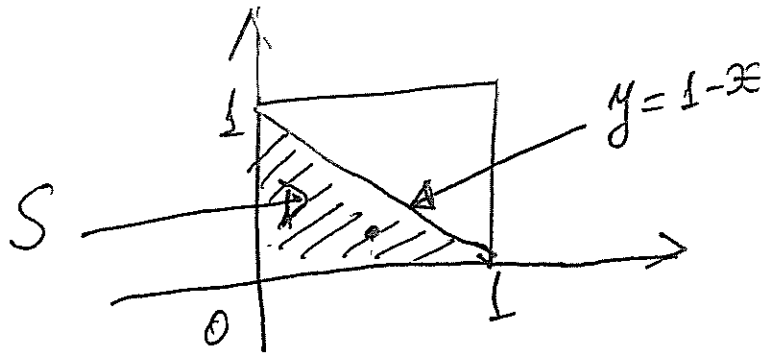
$$f_Y(y) = \begin{cases} 2y; & 0 \le y \le 1 \\ 0; & \text{otherwise.} \end{cases}$$

- Independence $\Rightarrow f(x,y) = f_X(x) \cdot f_X(y)$

$$= \begin{cases} 4xy; & 0 \le x \le 1 \\ & \text{and} \\ & 0 \le y \le 1 \\ 0; & \text{otherwise.} \end{cases}$$

- $\mathbb{P}(X+Y \le 1) = \iint f(x,y)\,dx\,dy.$

$$S = \{(x,y) \in \mathbb{R}^2 \mid x+y \le 1\}.$$



$$S = \{(x,y) \in \mathbb{R}^2 \mid 0 \le x \le 1,\ 0 \le y \le 1-x\}$$

Then:

$$\mathbb{P}(X+Y \le 1) = \int_0^1 \left(\int_0^{1-x} 4xy\,dy \right) dx = \frac{1}{6} //$$

How to check if variables are independent, given their joint p.d.f (p.f.)?

Note (Theorem 3.5.1)

X indep. of Y iff $\boxed{f(x,y) = h_1(x) \cdot h_2(y)}$ (*)

where h_1 is a fctn. of x alone,

and

h_2 is a fctn of y alone;

and (*) MUST hold for all x and y.

Example (**)

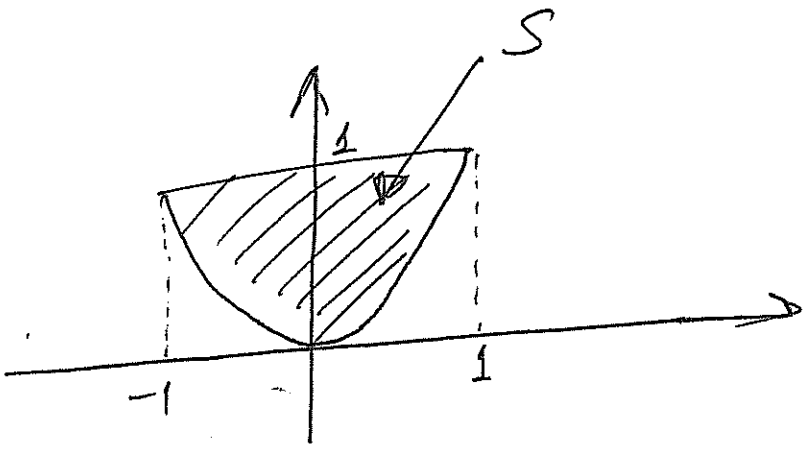
Recall that we considered this example:

$$(X,Y) \sim f(x,y) = \begin{cases} \frac{21}{4}x^2y, & \text{if } x^2 \le y \le 1, \\ 0, & \text{otherwise,} \end{cases}$$

and we found:

$$f_X(x) = \begin{cases} \left(\frac{21}{8}\right)x^2(1-x^4), & \text{for } -1 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} \left(\frac{7}{2}\right)y^{\frac{5}{2}}, & \text{for } 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$



$f(1,1) = \frac{21}{4} \neq f_X(1) \cdot f_Y(1) = 0$. X and Y are NOT independent.

Note The boundaries of the support S above are "curved": Information on x influences that on y.

Result If $S = \{(x,y) \mid f(x,y) > 0\}$ is a rectangle (where $f(x,y)$ is the pdf of (X,Y)), and its sides are parallel to the coordinate axes, then:

X and Y are independent iff (*) holds for all $(x,y) \in S$.

NB1 This result says that we only look at the expression of $f(x,y)$ for $(x,y) \in S$ and that we may NOT need to calculate the marginal distributions to check independence.

NB2 If $(X,Y) \sim f(x,y)$, for $(x,y) \in D$ (where D is the whole domain of f), if $X \sim f_X(x)$ and $Y \sim f_Y(y)$ and if you want to show that X and Y are NOT independent, it is enough to find only one point, call it (x_0, y_0), for which $f(x_0,y_0) \neq f_X(x_0) \cdot f_Y(y_0)$.

Example.

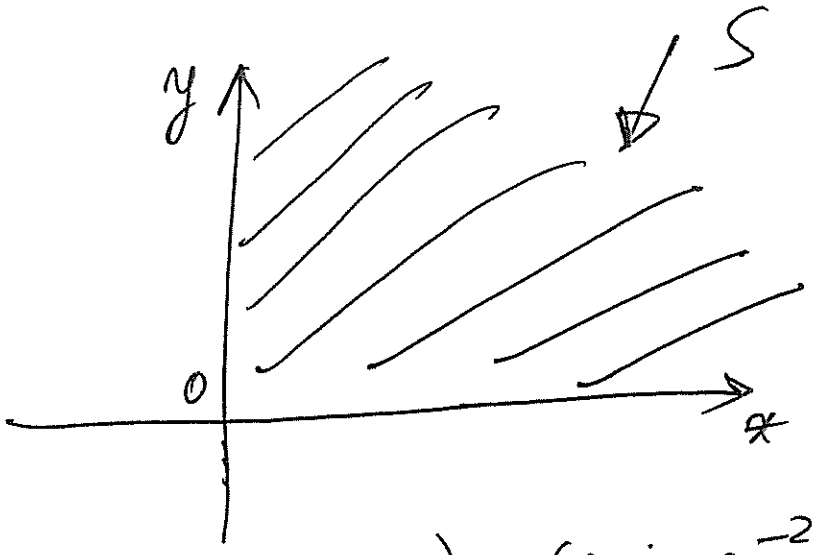
Let $(X,Y) \sim f(x,y) = \begin{cases} k \cdot e^{-(x+2y)}, & \text{if } x \geq 0,\ y \geq 0 \\ 0, & \text{otherwise,} \end{cases}$

where $k>0$ is some constant.

① Are X and Y independent?

② Find their respective marginals? (See textbook)

Solution



Clearly $f(x,y) = (k_1 \cdot e^{-x}) \cdot (k_2 \cdot e^{-2y})$,

for $k_1 \cdot k_2 = k$, for all $(x,y) \in S$, so

X and Y are independent.

Conditional distributions

Discrete conditional distributions

- Let (X, Y) have joint p.f. $f(x, y)$
- For every y s.t. $f_2(y) > 0$

$$g_1(x|y) = \frac{f(x,y)}{f_2(y)}$$ is called the conditional distribution of X given that Y took on value y (given that $Y = y$).

- Similarly, for every x s.t. $f_1(x) > 0$,

$$g_2(y|x) = \frac{f(x,y)}{f_1(x)}$$ is the conditional distribution of Y given that $X = x$.

Note $f_1(x)$ and $f_2(y)$ are the marginals of X and Y, respectively.

Result Consider $g_1(x|y)$. Fix the value of y. Then $g_1(x|y)$ is a p.f., viewed as a function of x:

$$\sum_x g_1(x|y) = \sum_x \frac{f(x,y)}{f_2(y)}$$

$$= \frac{\sum_x f(x,y)}{f_2(y)} = \frac{f_2(y)}{f_2(y)} = 1.$$

(The equality $\sum_x f(x,y) = f_2(y)$ is the definition of marginal distributions.)

Example $X = \begin{cases} 0, & \text{a car is not stolen} \\ 1, & \text{a car is stolen} \end{cases}$

$Y \in \{1, 2, 3, 4, 5\}$: 5 different car brands.

Table 3.7 Joint p.f. for Example 3.6.1

	Brand Y					
Stolen X	1	2	3	4	5	Total
0	0.129	0.298	0.161	0.280	0.108	0.976
1	0.010	0.010	0.001	0.002	0.001	0.024
Total	0.139	0.308	0.162	0.282	0.109	1.000

Table 3.8 Conditional p.f. of X given Y for Example 3.6.3

	Brand Y				
Stolen X	1	2	3	4	5
0	0.928	0.968	0.994	0.993	0.991
1	0.072	0.032	0.006	0.007	0.009

. Tables (3.7) and (3.8).

. We can verify that

$$P(X=1 \mid Y=1) = \frac{f(1,1)}{f_2(1)} = \frac{0.010}{0.139} \simeq 0.072$$

$$P(X=1 \mid Y=2) = \frac{f(1,2)}{f_2(2)} = \frac{0.010}{0.308} \simeq 0.032$$

and so on. Therefore, based on these conditional probabilities, it appears that Brand 1 is (much) more likely to be stolen. Contrast this with $P(X=1) = 0.024$, which is the (overall) probability of a car being stolen, irrespective of brand.

Continuous conditional distributions

Definition Let $(X, Y) \sim f(x,y)$, and let $f_1(x)$ and $f_2(y)$ be the marginals.

- For each point y s.t. $f_2(y) > 0$, the conditional pdf of X given $Y = y$ is:

$$g_1(x|y) = \frac{f(x,y)}{f_2(y)}, \text{ for } -\infty < x < \infty.$$

- For each point x s.t. $f_1(x) > 0$, the conditional pdf of Y given $X = x$ is:

$$g_2(y|x) = \frac{f(x,y)}{f_1(x)}, \text{ for } -\infty < y < \infty.$$

NOTE: For a given (fixed) y, $g_1(x|y)$ is a p.d.f. Similarly, for a given x, $g_2(y|x)$ is a pdf (as a fcn. of y).

Example

- Recall example (**) at page 52 above.
- Notice that $f_X(x) > 0$ for all $x \in (-1,1)$, $x \neq 0$. Therefore, for any such x we can define:

$$g_2(y|x) = \begin{cases} \frac{2y}{1-x^4}, & \text{for } x^2 \leq y \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

- How to use it?

$$\mathbb{P}\left(Y \geq \frac{3}{4} \,\middle|\, X = \frac{1}{2}\right) = \int_{\frac{3}{4}}^{1} g_2\left(y \,|\, x = \frac{1}{2}\right) dy$$

$$= \int_{\frac{3}{4}}^{1} \frac{2y}{1-\left(\frac{1}{2}\right)^4} dy = \frac{16}{15} \cdot 2 \int_{\frac{3}{4}}^{1} y\, dy$$

$$= \frac{32}{15} \cdot \left(\frac{y^2}{2} \Big|_{\frac{3}{4}}^{1} \right) = \frac{16}{15}\left[1 - \frac{9}{16}\right]$$

$$= \frac{16}{15} \cdot \frac{7}{16} = \frac{7}{15} \,//.$$

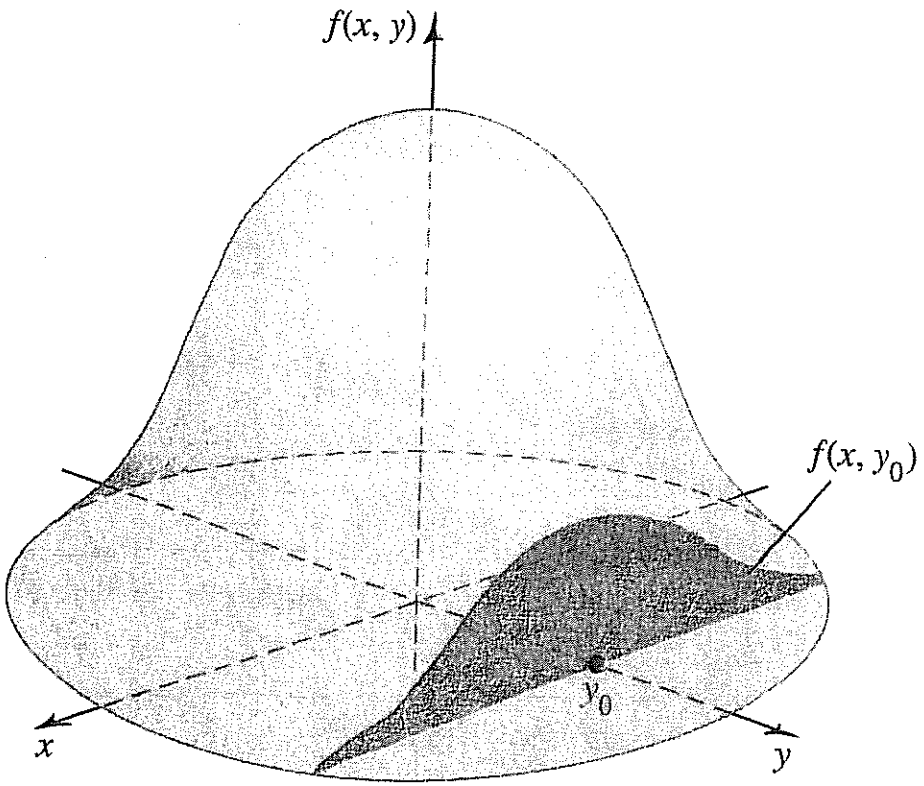


Figure 3.20 The conditional p.d.f. $g_1(x|y_0)$ is proportional to $f(x, y_0)$.

NOTE See Figure 3.20 to understand that when we write $\mathbb{P}(X \geq x_0 \mid Y = y_0)$ we do NOT condition on an event of probability zero (as $\{Y = y_0\}$ has zero probability if Y is continuous), but rather use a notation that says that we should work in a reduced sample space, where the reduction is obtained by looking only at pairs (x, y_0), with fixed y_0 and variable x, and use the resulting distribution:

$$g_1(x \mid y_0) = \frac{f(x, y_0)}{f_2(y_0)}$$

$\propto f(x, y_0)$, with proportionality factor $f_2(y_0)$.

"Proportional"

Construction of the Joint distribution from marginal and conditional pf/pdf's.

- **Multiplication rule**

$$f(x,y) = g_1(x|y) \cdot f_2(y), \quad (\forall) y \text{ s.t. } f_2(y) > 0.$$

or

$$f(x,y) = g_2(y|x) \cdot f_1(x), \quad (\forall) y \text{ s.t. } f_1(x) > 0$$

- **Law of total probability for r.v.'s**

(*) $f_1(x) = \sum_y g_1(x|y) \cdot f_2(y)$, if Y is discrete.

(**) $f_1(x) = \int_{-\infty}^{\infty} g_1(x|y) f_2(y) dy$, if Y is continuous.

Bayes theorem for random variables

(1) $g_2(y|x) = \dfrac{g_1(x|y)\, f_2(y)}{f_1(x)}$, with

$f_1(x)$ given by (*) or (**).

(2) $g_1(x|y) = \dfrac{g_2(y|x) \cdot f_1(x)}{f_2(y)}$, with

$f_2(y)$ given by what (*) or (**) become by switching x and y and the subscripts 1 and 2.

Theorem. (1) X and Y are independent iff $g_1(x|y) = f_1(x)$, for any x and any y s.t. $f_2(y) > 0$

(2) X and Y are independent iff $g_2(y|x) = f_2(y)$, for every y and every x s.t. $f_1(x) > 0$.

Example

- Choose X at random from $[0,1]$. That is, assume $X \sim \text{Unif}[0,1]$,

$$X \sim f_1(x) = \begin{cases} 1, & \text{if } 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- After you chose a value, say $x \in [0,1]$ of X, choose now Y, uniformly on $[x,1]$. This means that we are given the conditional distribution of Y, for each x:

$$g_2(y|x) = \begin{cases} \frac{1}{1-x}, & \text{for } x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, by the multiplication rule:

$$f(x,y) = \begin{cases} \frac{1}{1-x}, & \text{for } 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

• The marginal distribution of Y is

$$f_2(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_0^y \frac{1}{1-x}\,dx$$

$$= -\int_1^{1-y} \frac{du}{u}$$

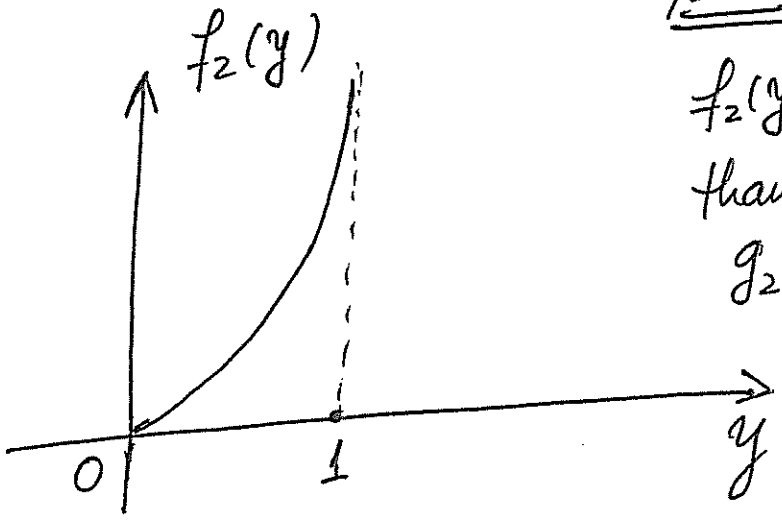
$1-x = u \Rightarrow dx = -du$

$x = 0 \Rightarrow u = 1$

$x = y \Rightarrow u = 1-y$

$$= \int_{1-y}^{1} \frac{du}{u} = \ln u \Big|_{u=1-y}^{u=1}$$

$$= \ln 1 - \ln(1-y) = -\ln(1-y).$$



NOTE The marginal $f_2(y)$ looks different than the conditional $g_2(y|x)$!